MODULE 4

Systems & Projections

Linear Algebra III — the machinery behind fitting a model to data

Mathematical Foundations for Data Science & Computational Intelligence · Mohsin Raza · Fall 2026

This week: two sessions, one question

SESSION 1

When the equations disagree

- When does Ax = b have a solution?
- What 'best' means when it doesn't
- Least squares as a projection
- The normal equations, by hand

SESSION 2

Computing it without breaking it

- Why the textbook formula fails on real data
- QR: Gram–Schmidt, done properly
- SVD & the pseudoinverse (Module 3 returns)
- Which solver to trust, and when

you are here — and Module 7 (Regression) is built directly on it

SESSION 1

When the equations disagree

Two measurements, two unknowns

Every request to a web service costs:

- a fixed overhead (connect, authenticate, start up) — paid once, whatever the size
- a cost per MB — more data, proportionally more time

$$t = x_1 + x_2 \cdot s$$

x_1 = overhead (ms)
x_2 = cost per MB (ms/MB)

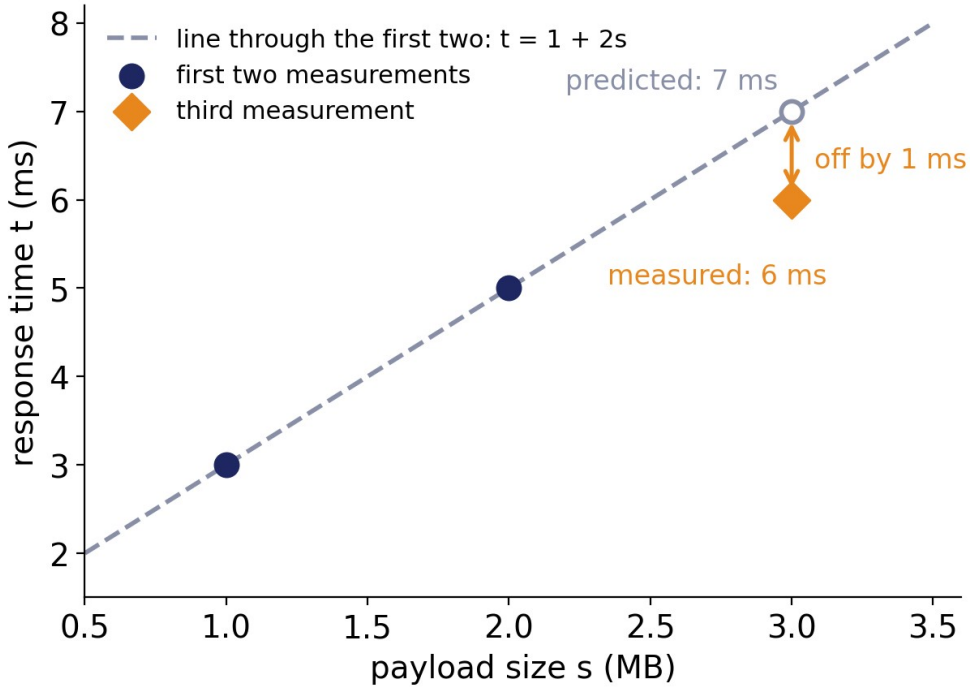
payload s	time t	equation
1 MB	3 ms	$3 = x_1 + 1 \cdot x_2$
2 MB	5 ms	$5 = x_1 + 2 \cdot x_2$

The x's are the unknowns we solve for. The data (s and t) are what we measured.

?

Two equations, two unknowns. What are the overhead and the cost per MB?

Then a third measurement arrives



1 Which measurement is wrong?

2 Should we throw one away? Which one?

3 If none is 'wrong' — what should the answer even mean now?

“Least wrong” — measured how?

error = measured time − predicted time. We want one number that scores a whole line.

Add up the errors

Positive and negative errors cancel. The flat line t = 4.67 ignores payload size, yet its errors −1.67, +0.33, +1.33 add up to exactly 0: a “perfect” score.

Add up |errors|

Reasonable, and gentler on one wild outlier. But it measures distance like walking city blocks, so there's no clean formula — you have to search. (Module 7's Lasso uses this idea.)

Add up errors^2

Stack the errors into one vector: this is its squared straight-line length, $\|Ax - b\|^2$ from Module 2. Smallest error = closest point. That's geometry.

Least squares:

$$\hat{\mathbf{x}} = \arg\min_{\mathbf{x}} \ \|A\mathbf{x} - \mathbf{b}\|^2$$

The column picture of Ax = b

$$A\mathbf{x} = \mathbf{b}: \quad x_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \stackrel{?}{=} \begin{bmatrix} 3 \\ 5 \\ 6 \end{bmatrix}$$

Ax is always a combination of A's columns.
So Ax = b has a solution exactly when b lies in the column space, col(A).

Here: two columns in 3-D space span a plane. The data vector (3, 5, 6) sticks out of that plane.

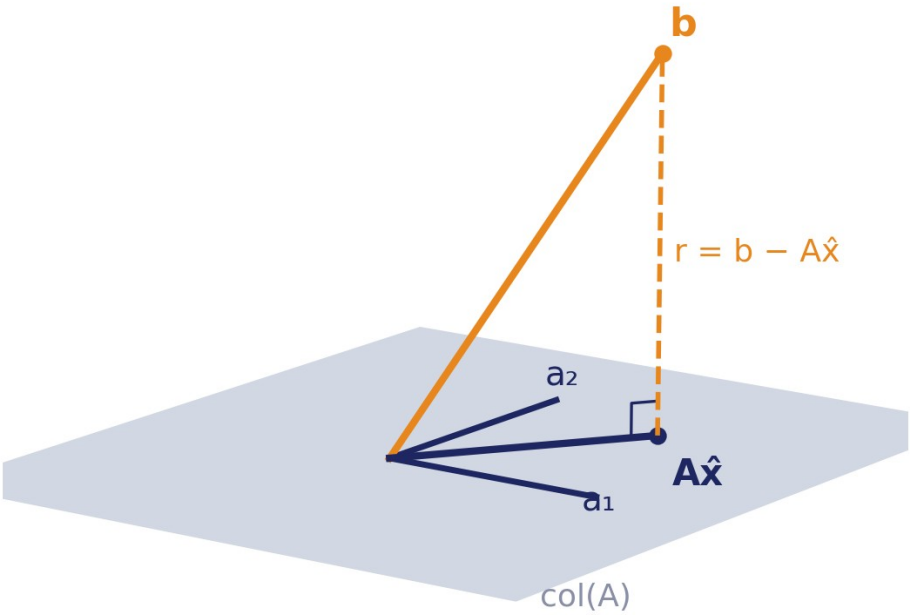
Rank test (Module 2): rank(A) = 2, rank([A | b]) = 3. Adding b as a column raised the rank — so b is not a combination of A's columns. No exact solution exists.

Every linear system lands in one of three cases

Shape / rank	Solutions	Where you meet it
Square, full rank	**Exactly one**	Textbook exercises; rare with real data
Tall: more equations than unknowns	**Usually none**	Classic regression — many data points, few parameters
Wide, or redundant columns	**Infinitely many**	Duplicated features; overparameterized models

Session 1 answers the tall case (least squares). Session 2 handles the redundant case with the SVD.

The best fit is a shadow



No x reaches b exactly. So find the point of col(A) that is **closest** to b.

Which point on the floor is closest to a light bulb hanging above it?

Module 2: projecting onto a line. Today: onto a plane — the same idea, one more dimension.

Perpendicular residual ⇒ normal equations

$$A^\top(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0} \quad \Longrightarrow \quad A^\top A\,\hat{\mathbf{x}} = A^\top \mathbf{b}$$

1

The residual r = b − A$\hat{x}$ must be perpendicular to the plane

2

Perpendicular to the plane ⇔ perpendicular to every column of A ⇔ A^Tr = 0

3

Rearrange: a small square system — 2 × 2 here, no matter how many data points

By hand, on our three measurements

$$A^\top A = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}, \quad A^\top \mathbf{b} = \begin{bmatrix} 14 \\ 31 \end{bmatrix} \quad \Longrightarrow \quad \hat{\mathbf{x}} = \begin{bmatrix} 5/3 \\ 3/2 \end{bmatrix}$$

1.67 ms

fixed overhead

1.5 ms

per MB

0

sum of the residuals (−0.17, +0.33, −0.17)

Why do the residuals sum to exactly zero?

You've been running this for years

Every one of these is solving the same projection:

```
lm(dist ~ speed)
Excel “Add Trendline”
numpy.linalg.lstsq(A, b)
sklearn LinearRegression().fit(X, y)
```

The projection itself is a matrix:

$$P = A\,(A^\top A)^{-1} A^\top, \qquad P^2 = P, \qquad P^\top = P$$

Project twice = project once ($P^2 = P$).

Statisticians call P the “hat matrix” — it puts the hat on y: $\hat{y} = Py$. You'll meet it again in Module 7's leverage and diagnostics.

BEFORE NEXT SESSION

Every textbook writes the answer as $(A^TA)^{-1}A^Tb$.

R's lm() never computes it. Why not?

Questions?